

SuryaWatch

Plan it. Then watch it.

User guide and feature overview · 9 October 2026

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1. What SuryaWatch is

SuryaWatch is a free web app for Indian homes and rooftop solar. It does two jobs with one solar engine:

- **Plan:** for a home without solar. Mark the roof on a satellite map and enter the electricity bill. SuryaWatch gives the right system size, the cost after PM Surya Ghar and Delhi subsidies, the monthly benefit, the payback period and the CO2 avoided.
- **Watch:** for a home that already has solar. Photograph the inverter display. AI reads the numbers, SuryaWatch compares them with what that day's sunlight should have produced, and says whether a shortfall is **smog** (not your fault), **dust** (clean the panels, with the rupees lost per week) or a **fault** (call your installer).

“Plan promises you a number of units a year. Watch makes sure you actually get them.”

The problem it solves

- Rooftop solar is growing fast: PM Surya Ghar crossed 50 lakh homes in August 2026. Most new owners never check whether their panels are producing what they should.
- Dust and air pollution can cut Indian solar output by 17–25% (Duke University study at IIT Gandhinagar). Delhi's October–November smog is the worst case. Owners cannot tell smog, dust and a fault apart from an inverter number alone.
- Many families stall before installing: over 1 crore had registered on the PM Surya Ghar portal by May 2026, but far fewer had installed. Sizing, cost and paperwork are the usual blockers.

2. Features at a glance

Feature	What it does	Runs on
Roof planner	Draw the roof on a satellite map; get system size, yearly units, cost, subsidies, monthly benefit, payback, 25-year benefit and CO2 avoided.	AWS Lambda, NASA POWER data
Subsidy maths	PM Surya Ghar central subsidy, Delhi capital subsidy and generation incentive, DERC slab tariff and the Delhi power subsidy, net metering.	AWS Lambda
Photo reading	Upload inverter-display photos (several at once). AI reads power, today's energy, total energy and the inverter state. You check the numbers before saving.	Amazon Bedrock (Amazon Nova)
Daily verdict	Expected output from that day's real sunlight, temperature and haze; splits any shortfall into sky loss (smog) and panel loss (dust or fault); rupees lost per week.	AWS Lambda, Open-Meteo data
Fault codes	If the State line on the display shows Fault or an error code, the day is flagged at once.	AWS Lambda
Charts	Energy made so far vs what the sunlight allowed; power readings vs expected power.	Web app
Day-by-day calendar	A green square per day (darker = better). Click a day to open it.	Amazon DynamoDB
Cleaning effect	Log a cleaning; SuryaWatch compares the days before and after (e.g. +27%, ₹122 a week).	Amazon DynamoDB
Email alerts	Every evening at 7:30 PM: dust, fault, smog day or a missing reading, with a one-line Hindi summary.	Amazon EventBridge Scheduler, Amazon SNS
Ask SuryaWatch	Questions in English or Hindi, answered with your own numbers and the scheme rules.	Amazon Bedrock
Share link	Each rooftop has its own link, so the whole family can open the same dashboard.	Web app

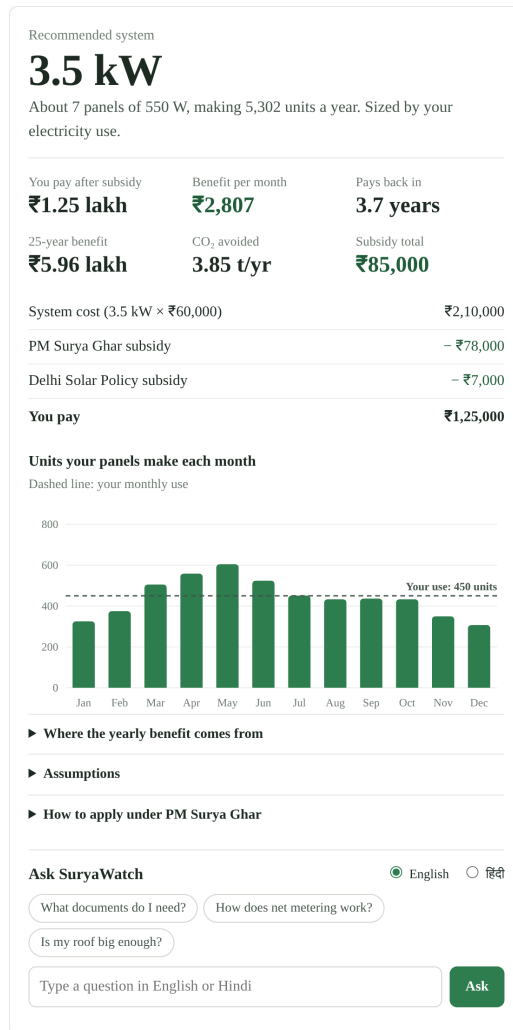
3. How to use: Plan mode (should my home go solar?)

Open the app. The **Plan** tab opens first.

1. **Find your roof.** Search your address or press *My location*, then click your house on the satellite map.
2. **Draw the roof.** Press *Draw roof outline* and click each corner of the roof. Double-click or press *Finish outline*. The area fills in by itself; you can also type it in square feet or square metres.
3. **Usable share.** Leave out the water tank, stair room and edges. Most roofs are 60–80% usable.
4. **Your electricity.** Enter monthly units (from the bill) or the monthly bill in rupees, and choose Delhi or Other state.
5. Press **Plan my solar**.

Reading the result

- **Recommended system:** size in kW and number of panels, limited by your roof or your use.
- **You pay after subsidy, benefit per month, pays back in, 25-year benefit, CO2 avoided.**
- **Cost breakdown:** system cost, PM Surya Ghar subsidy, Delhi subsidy, what you pay.
- **Monthly chart:** units your panels make each month against your use (dashed line).
- Open *Assumptions* to see every number used, and *How to apply* for the 9 portal steps.
- Use **Ask SuryaWatch** under the result for questions in English or Hindi.



Plan result for a Delhi home using 450 units a month (screenshot, sample inputs).

4. How to use: Watch mode (is my solar working?)

4.1 Set up your rooftop (once)

1. Open the **Watch** tab.
2. Enter a name, the location (long-press the building in Google Maps and paste the two numbers, or press *Use my location*), the system size in kW, which way the panels face, and the tilt.
3. Press **Start watching**. This device remembers the rooftop. *Copy link* gives a link to share; on another device paste that link or the system ID into *Already set up?*

1 Set up your rooftop

Once per system. This device remembers it, and you can share the link with your family.

Name

Our rooftop

Location (latitude, longitude)

28.6139, 77.2090

Use location from Plan

Use my location

Or long-press your building in Google Maps and paste the numbers.

System size (kW)

3

Panels face

South

Tilt (degrees)

20

► Advanced

Start watching

Already set up? Paste your SuryaWatch link or system ID

e.g. cc2572e846

Open

The one-time setup form.

4.2 Add readings

1. Press **Upload display photos** and pick one or more photos of the inverter screen. The time each photo was taken is read from the photo itself.
2. AI reads *Power now*, *Today (kWh)*, *Total (kWh)* and the *Inverter state*. Check the numbers, correct anything wrong, then press **Save**.
3. No photo? Press **Type a reading** and enter the numbers yourself.
4. Wrong reading? Use *Remove* in the readings table at the bottom.

Add readings

Photograph every screen the inverter shows: power now, today's energy (E-Today) and total energy. Upload original photos, not WhatsApp copies, so the time they were taken comes through.

Upload display photos

Type a reading



Time

10/09/2026, 01:05 P

Power now (kW)

1.67

[Discard](#)

Today (kWh)

8.2

Total (kWh)

5665

Inverter state

Normal

Read by AI (medium confidence). Check the numbers before saving. TEST MODE: simulated reading, not from the photo.

Save 1 reading

Reviewing a photo of the Eastman EGT-3K-1P1M display before saving. On a laptop the AI numbers are simulated (test mode); after deployment Amazon Bedrock reads the photo.

Taking good display photos (Eastman SolarLink EGT-3K-1P1M and similar)

- Press ESC or ▲ once to wake the screen.
- Block the glare with your body; no flash. Hold the phone straight, 20–30 cm away, and tap the screen on your phone to focus.
- Press ▼ through the screens and photograph each one: Power and State, E-Today (or Day) and E-Total (or Total).
- Take the last photo of the day at about **5:45 PM**, before the inverter switches off for the night. SuryaWatch counts it as the day's total.
- Share photos through Google Drive, or on WhatsApp **as a Document**. Normal WhatsApp photos lose the time they were taken.

4.3 Read the day's verdict

The verdict card answers one question: *did the panels make what the day's sunlight allowed, and if not, why?*

- **Made:** your inverter's energy for the day.
- **Sunlight allowed:** what a clean system would make under that day's real sunlight and heat.
- **Performance:** made ÷ allowed.
- **Haze cut sunlight:** extra loss from smog, estimated from that day's aerosol readings.
- **Panel loss:** what is left after haze: dust, shade or a fault.
- **Lost per week:** panel loss in rupees if it continues.

Clean panels

Dust on panels - clean them

After allowing for haze, your panels made 21% less than they should. That is about ₹120 a week. Clean early morning or evening with plain water and a soft cloth.

Made	Sunlight allowed	Performance
10.7 kWh	13.74 kWh	78%
Haze cut sunlight	Panel loss	Lost per week
1%	21%	₹120

That day a 3 kW system here should make about 13.7 kWh in total. Sunlight 86% of a clear day. Air: PM2.5 70 µg/m³, aerosol depth 0.45.

Ask SuryaWatch

☐ English
☒ हिंदी

आज बिजली कम क्यों बनी?

पैनल सुरक्षित तरीके से कैसे साफ करें?

इस नतीजे का मतलब क्या है?

A dusty day in the sample data.

Verdict	What it means	What to do
Working well	Output is at least 92% of what the sunlight allowed.	Nothing.
Smog day	Haze blocked sunlight; the panels made what the sky allowed.	Nothing to fix.
Clean panels	Panel loss after allowing for haze, with no recent rain or cleaning.	Clean early morning or evening with plain water and a soft cloth. If rain is likely in the next two days, SuryaWatch says so: wait for it and save the water.
Possible fault	Sudden drop against recent days, or the display shows Fault or an error code.	Note the code, check the inverter, call the installer.
Check system	Output is low even though the panels were cleaned or rained on recently.	Look for new shade, a loose cable or inverter warnings.
No reading / Too early	No energy reading yet for the day, or it is before about 10 AM.	Add a reading later in the day.

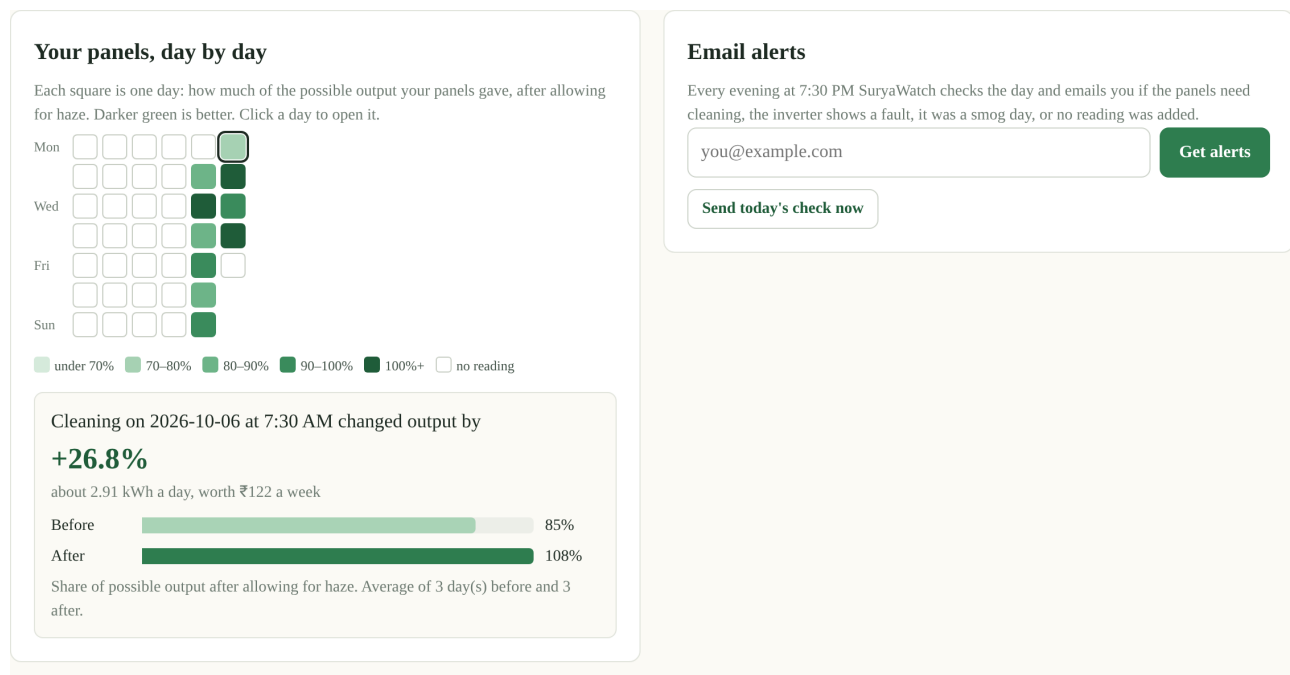
4.4 Charts



Top: energy made so far (dots) against what the sunlight allowed (line). Bottom: power readings against expected power. Hover for values. Sample data.

Use the < > buttons or the date box at the top of the dashboard to look at other days.

4.5 Day-by-day calendar, cleaning effect and alerts



Left: the green calendar and the cleaning effect. Right: email alerts. Sample data.

- **Calendar:** each square is one day; darker green means more of the possible output (after haze). A red outline marks a fault. Click any square to open that day.
- **Cleaning effect:** after cleaning, press *We cleaned the panels today*. Once there are full days on both sides, SuryaWatch shows the change in output and its value per week.
- **Email alerts:** enter an email and press *Get alerts*. After deployment, AWS sends a confirmation email: click the link once. Every evening at 7:30 PM you then get an email when there is dust, a fault, a smog day, or no reading. *Send today's check now* runs the

check at once.

4.6 Ask SuryaWatch (English or Hindi)

Under every Plan result and every daily verdict there is a helper. Choose English or Hindi, tap a ready-made question or type your own. It answers in about 120 words using your own numbers and the PM Surya Ghar and Delhi Solar Policy rules, and says so when it is not sure.

5. How SuryaWatch works

1. **Sun position and clear-sky sunlight** for any place and minute (NOAA equations, Haurwitz model).
2. **Real sunlight on the panel** for each hour from Open-Meteo, for the panel's tilt and direction, plus temperature, cloud and rain.
3. **Expected output** = system kW × sunlight on the panel ÷ 1000 × 0.85 (a clean system's performance ratio) × heat loss (−0.35% per °C of panel temperature above 25 °C).
4. **Haze**: weather models only know average haze, so SuryaWatch estimates the day's extra haze loss from aerosol optical depth (Open-Meteo air quality): about $0.25 \times (\text{AOD} - 0.4)$, capped at 40%.
5. **Verdict**: the rest of the gap is panel loss. A gradual loss with no recent rain or cleaning is dust; a sudden drop or an error code is a fault.
6. **Plan**: long-term monthly sunlight (NASA POWER) → units per kW per year (about 1,500 in Delhi) → size limited by roof (10 m² per kW) and by use → bills before and after → subsidies, incentive and payback with 0.5% panel ageing a year.

6. Built on AWS

AWS service	What it does in SuryaWatch
AWS Amplify Hosting	Hosts the web app.
Amazon API Gateway (HTTP API)	One entry point for every request from the app.
AWS Lambda (Python)	Plan maths, expected output, verdicts, photo handling, alerts.
Amazon Bedrock (Amazon Nova 2 Lite)	Reads inverter photos; answers questions in English and Hindi; Hindi line in alert emails.
Amazon DynamoDB	Rooftops, readings, daily verdicts, cleanings, alert settings.
Amazon S3	Stores the inverter photos (private).
Amazon EventBridge Scheduler	Runs the evening check every day at 7:30 PM IST.
Amazon SNS	Sends the alert emails.
AWS SAM (CloudFormation)	Defines all of the above in one file, template.yaml, deployed with one script.

7. Running SuryaWatch on your laptop

Open a terminal (in VS Code) inside the *suryawatch* folder. You need Python 3.12 and Git; nothing else.

Command	What it does
<code>python -m unittest discover -s tests</code>	Runs all automated tests (should end with OK).
<code>python scripts/local_server.py --mock-ai</code>	Starts the app at <code>http://localhost:8080</code> with live weather and simulated AI (test mode). Keep this terminal open; press Ctrl+C to stop.
<code>python scripts/local_server.py --offline --mock-ai</code>	Same, with sample weather; no internet needed.
<code>python scripts/sample_data.py</code>	In a second terminal: creates the “Sample rooftop (demo data)” with 10 days of generated readings and prints its link. Takes about a minute.
<code>python scripts/import_csv.py --system ID file.csv</code>	Adds real readings from a spreadsheet (columns: time, power_kw, e_today_kwh, e_total_kwh, state).
<code>python backend/daily_check.py</code>	Runs the evening check by hand. On a laptop, alerts are written to <code>data/outbox/alerts.log</code> .
<code>git add -A, git commit -m "...", git push</code>	Saves your work to GitHub.

On the laptop, data is kept in the *data* folder (Git ignores it). Real AI photo reading, the AI helper and real emails need AWS, so they switch on after deployment. Add `--api <deployed API address>` to `sample_data.py` or `import_csv.py` to send data to the deployed app instead.

8. Limits and honest notes

- All rupee figures are estimates. PPAC, fixed charges and tax are not included, and the installed cost (₹60,000 per kW by default) can be changed under *Advanced assumptions*. Get vendor quotes before buying.
- The haze estimate is a transparent rule of thumb, not a measurement. The cleaning test on a real roof is how we check it.
- The AI can misread a faint or glary display. That is why every AI reading is shown for checking before it is saved, and unreadable digits are left blank rather than guessed.
- The sample rooftop's readings are generated, not measured, and it is labelled “demo data”. Real readings come from the team's rooftop and replace it in the demo.
- Code and documentation were written with Claude (Anthropic) as the AI coding assistant, directed and tested by the team, as the hackathon rules require us to disclose.

9. Sources

Environmental Hacks rules and tracks: wemakedevs.org/aws/env · PM Surya Ghar 50 lakh homes: YourStory, Aug 2026 · 1 crore+ registered, 33 lakh installed: Energetica India, May 2026 · Central subsidy slabs and portal steps: PIB, Mar 2025 · Delhi Solar Policy 2024: Ornate Solar summary · DERC domestic tariff slabs: Bridgeway Power, 2026 · Grid emission factor 0.727 kg CO₂/kWh: CEA via SolarQuarter, FY 2023-24 · Air pollution cuts solar output 17-25%: Duke University study, reported by CleanTechnica · Weather and air quality: Open-Meteo · Long-term sunlight: NASA POWER · Amazon Nova 2 Lite: AWS Bedrock documentation.